ESE 5700: Digital Integrated Circuits and VLSI Fundamentals

Project 1, Part 1: The 6T Bitcell

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Team and contributions

Member	Contributions (this part)
Tyrone Marhguy	*TBD*
Victor Wanjohi	*TBD*

Both team members are responsible for understanding the full design and report.

1. Sizing (β and γ)

Hand analysis

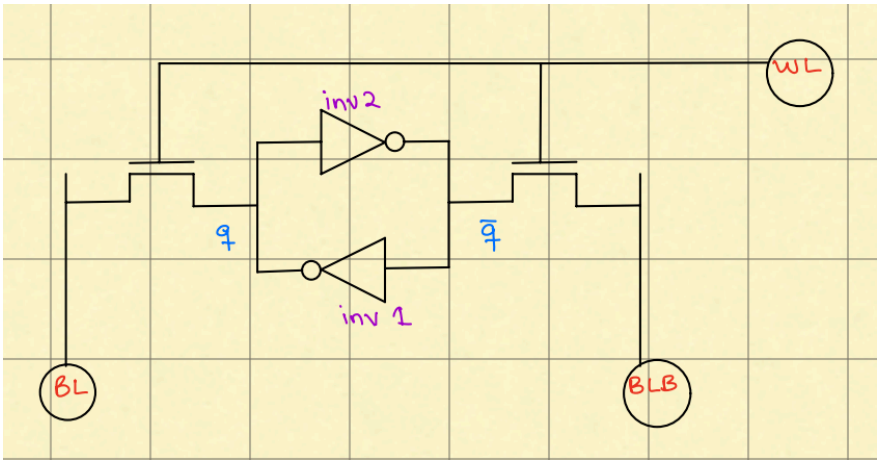


Figure 1: 6T cell read-disturb setup: $Q = 0$ stored, wordline asserted, bitlines precharged (hand sketch).

Read stability

Assume the stored state $Q = 0$ ($\overline{Q} \approx V_{DD}$) for read-stability analysis. With $Q = 0$ the "0" node is pulled down by the NMOS inside INV1 while the bitline (precharged to V_{DD}) drives current through the pass-gate, so the disturbed node follows an approximate voltage-divider model:

$$\frac{V_Q}{V_{DD}} = \frac{R_{\mathrm{PD1}}}{R_{\mathrm{PG}} + R_{\mathrm{PD1}}} = \frac{1}{1 + R_{\mathrm{PG}}/R_{\mathrm{PD1}}}. \tag{1}$$

From transistor physics $R_T \propto 1/W_T$ at fixed L. Defining the cell ratio $\beta = (W/L)_{\mathrm{PD}}/(W/L)_{\mathrm{PG}} = W_{\mathrm{PD}}/W_{\mathrm{PG}}$ gives $\beta \approx R_{\mathrm{PG}}/R_{\mathrm{PD}}$, hence

$$\frac{V_Q}{V_{DD}} \approx \frac{1}{1+\beta}, \qquad V_Q \approx \frac{V_{DD}}{1+\beta} : \tag{2}$$

a larger β lowers the read-disturb bump. Whether that bump flips the cell is set by the inverter switching voltage V_M ($V_{\mathrm{in}} = V_{\mathrm{out}}$), where both devices conduct.

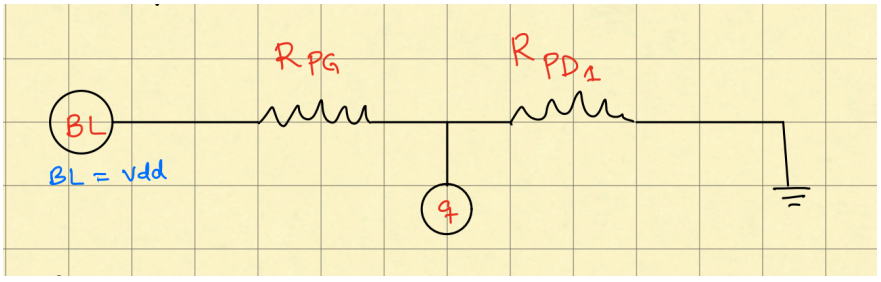


Figure 2: Approximate resistive divider for the read disturb: precharged bitline through R_{PG} against the pull-down R_{PD1} to ground (hand sketch).

Current balance at the equilibrium disturb voltage: neglecting gate leakage, KCL gives $I_{\mathrm{PD}} = I_{\mathrm{PG}}$. The pass-gate has $D = G = V_{DD}$, $S = V_Q$, so $V_{DS} = V_{GS} = V_{DD} - V_Q$ and $V_{DS} > V_{GS} - V_{Tn}$ always: saturation, with $I_{\mathrm{PG}} = \frac{1}{2}k_{\mathrm{PG}}(V_{GS} - V_{Tn})^2$ and $k_{\mathrm{PG}} = \mu_n C_{ox}(W/L)_{\mathrm{PG}}$. The pull-down (pd1) has $V_D = V_Q$, $V_G \approx V_{DD}$, $V_S = 0$, so $V_{DS} = V_Q < V_{GS} - V_{Tn}$: triode, with $I_{\mathrm{PD}} = k_{\mathrm{PD}}[(V_{GS} - V_{Tn})V_{DS} - V_{DS}^2/2]$. Equating the two with $E = V_{DD} - V_{Tn}$ yields

$$V_Q^2(1+\beta) + V_Q(-2E - 2E\beta) + E^2 = 0 \quad \Rightarrow \quad V_Q = E\left(1 \pm \sqrt{\frac{\beta}{1+\beta}}\right), \tag{3}$$

of which the positive root exceeds $V_{DD} - V_{Tn}$ and is inconsistent with the assumed saturated pass-gate / triode pull-down, leaving

$$V_Q = (V_{DD} - V_{Tn})\left(1 - \sqrt{\frac{\beta}{1+\beta}}\right). \tag{4}$$

This assumes ideal long-channel square-law behavior, equal NMOS thresholds, and negligible body effect.

The disturbed node Q drives INV2, whose switching point satisfies $I_{DN} = |I_{DP}|$ with both devices in saturation:

$$V_M = \frac{V_{Tn} + B\,(V_{DD} - |V_{Tp}|)}{1+B}, \qquad B = \sqrt{\frac{k_P}{k_N}} = \sqrt{\frac{\gamma\,\mu_p}{\beta\,\mu_n}}, \tag{5}$$

Here k_P and k_N are the pull-up and pull-down transconductance parameters. With β defined as PD/PG and γ as PU/PG, $k_P/k_N = \gamma\mu_p/(\beta\mu_n)$. Thus V_M depends on the supply, threshold voltages, and relative PMOS/NMOS strengths. Read stability requires $V_Q < V_M$:

$$(V_{DD} - V_{Tn})\left(1 - \sqrt{\frac{\beta}{1+\beta}}\right) < \frac{V_{Tn} + \sqrt{\frac{\mu_p\gamma}{\mu_n\beta}}\,(V_{DD} - |V_{Tp}|)}{1 + \sqrt{\frac{\mu_p\gamma}{\mu_n\beta}}}. \tag{6}$$

Increasing β strengthens the pull-down relative to the pass-gate and lowers the disturb voltage, but β also moves V_M and costs area: the final cell ratio is not selected by minimizing V_Q alone.

Write stability

Assume the stored state $Q = 1$ ($\overline{Q} \approx 0$) for the write analysis, with a 0 written through BL. With BL driven to 0 V and WL at V_{DD}, the pass-gate pulls the "1" node down while the PMOS inside INV1, still on because $\overline{Q} \approx 0$, pulls it up to V_{DD}. The written node therefore follows an approximate voltage-divider model:

$$\frac{V_Q}{V_{DD}} = \frac{R_{\mathrm{PG}}}{R_{\mathrm{PG}} + R_{\mathrm{PU1}}} = \frac{1}{1 + R_{\mathrm{PU1}}/R_{\mathrm{PG}}}. \tag{7}$$

From transistor physics $R_T \propto 1/(\mu_T W_T)$ at fixed L. The two devices are of opposite type, so the mobility ratio does not cancel as it does in the read divider. Defining the pull-up ratio $\gamma = (W/L)_{\mathrm{PU}}/(W/L)_{\mathrm{PG}} = W_{\mathrm{PU}}/W_{\mathrm{PG}}$ gives $R_{\mathrm{PU1}}/R_{\mathrm{PG}} \approx \mu_n/(\mu_p\gamma)$, hence

$$\frac{V_Q}{V_{DD}} \approx \frac{1}{1 + \dfrac{\mu_n}{\mu_p\gamma}}, \qquad V_Q \approx V_{DD}\,\frac{\mu_p\gamma}{\mu_n + \mu_p\gamma}\,: \tag{8}$$

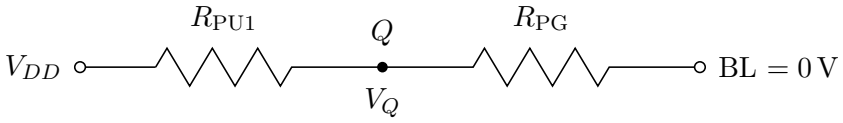


Figure 3: Approximate resistive divider for the write: supply through R_{PU1} against the pass-gate R_{PG} to the 0 V bitline.

a smaller γ lowers the written node. Whether it falls far enough to flip the cell is set by the switching voltage V_M ($V_{\mathrm{in}} = V_{\mathrm{out}}$) of INV2, which Q drives.

Current balance at the equilibrium write voltage: neglecting gate leakage, KCL gives $I_{\mathrm{PU}} = I_{\mathrm{PG}}$. The pass-gate has $G = V_{DD}$, $S = \mathrm{BL} = 0$ and $D = Q$, so $V_{GS} = V_{DD}$ and $V_{DS} = V_Q < V_{GS} - V_{Tn}$ at the equilibrium point: triode, with $I_{\mathrm{PG}} = k_{\mathrm{PG}}\big[(V_{DD} - V_{Tn})V_Q - V_Q^2/2\big]$ and $k_{\mathrm{PG}} = \mu_n C_{ox}(W/L)_{\mathrm{PG}}$. The pull-up (pu1) has $S = V_{DD}$, $G = \overline{Q} \approx 0$ and $D = Q$, so $V_{SG} = V_{DD}$ and $V_{SD} = V_{DD} - V_Q > V_{SG} - |V_{Tp}|$ as long as $V_Q < |V_{Tp}|$: saturation, with $I_{\mathrm{PU}} = \frac{1}{2}k_{\mathrm{PU}}(V_{DD} - |V_{Tp}|)^2$ and $k_{\mathrm{PU}} = \mu_p C_{ox}(W/L)_{\mathrm{PU}}$. Equating the two with $E = V_{DD} - V_{Tn}$, $F = V_{DD} - |V_{Tp}|$ and $k_{\mathrm{PU}}/k_{\mathrm{PG}} = \gamma\,\mu_p/\mu_n$ yields

$$V_Q^2 - 2E\,V_Q + \gamma\frac{\mu_p}{\mu_n}F^2 = 0 \quad \Rightarrow \quad V_Q = E \pm \sqrt{E^2 - \gamma\frac{\mu_p}{\mu_n}F^2}, \tag{9}$$

of which the positive root exceeds $V_{DD} - V_{Tn}$ and is inconsistent with the assumed triode pass-gate, leaving

$$V_Q = (V_{DD} - V_{Tn}) - \sqrt{(V_{DD} - V_{Tn})^2 - \gamma\frac{\mu_p}{\mu_n}\big(V_{DD} - |V_{Tp}|\big)^2}. \tag{10}$$

This assumes ideal long-channel square-law behavior, a saturated pull-up ($V_Q < |V_{Tp}|$), an ideal 0 V bitline, and no help from PG2 pulling $\overline{Q}$ up.

The written node Q drives INV2, whose switching point V_M is the one in Eq. (5), with $B = \sqrt{\gamma\mu_p/(\beta\mu_n)}$. Write stability requires $V_Q < V_M$:

$$(V_{DD} - V_{Tn}) - \sqrt{(V_{DD} - V_{Tn})^2 - \gamma\frac{\mu_p}{\mu_n}\big(V_{DD} - |V_{Tp}|\big)^2} \;<\; \frac{V_{Tn} + \sqrt{\frac{\mu_p\gamma}{\mu_n\beta}}\,(V_{DD} - |V_{Tp}|)}{1 + \sqrt{\frac{\mu_p\gamma}{\mu_n\beta}}}. \tag{11}$$

Equation (10) holds only while the pull-up is saturated, $V_Q < |V_{Tp}|$. With the values used here V_M is about 0.37 V, above $|V_{Tp}| = 0.3$ V, so (11) cannot be solved for γ with (10) alone. Requiring the stricter $V_Q < V_{Tn}$ keeps (10) valid when $V_{Tn} \le |V_{Tp}|$, guarantees that the NMOS of INV2 is off, and gives the required pull-up ratio in closed form:

$$\gamma < \frac{\mu_n}{\mu_p} \cdot \frac{V_{Tn}(2V_{DD} - 3V_{Tn})}{\big(V_{DD} - |V_{Tp}|\big)^2}. \tag{12}$$

For an illustrative calculation at $V_{DD} = 0.8$ V, with the assumed values $V_{Tn} = |V_{Tp}| = 0.3$ V and $\mu_n/\mu_p = 2$, (12) gives $\gamma < 1.68$. Evaluating (11) numerically at $\beta = 2$, with the triode pull-up current beyond $V_Q = |V_{Tp}|$, relaxes the limit to about $\gamma < 1.9$. For $\gamma = 1$ the divider (8) puts V_Q at 0.27 V and the current balance (10) at 0.15 V; the divider is pessimistic because a saturated pull-up supplies less current than a resistor would.

Table 1: Written-node voltage against pull-up ratio at $V_{DD} = 0.8\,\mathrm{V}$, $\mu_n/\mu_p = 2$.

γ	V_Q from (10) (V)	Margin to V_{Tn} (V)	Margin to V_M at $\beta = 2$ (V)
0.5	0.067	0.233	0.285
1	0.146	0.154	0.220
1.5	0.250	0.050	0.126
2	0.40, pull-up in triode	fails	fails

The bound scales directly with the NMOS-to-PMOS strength ratio. The assumed ratio must be replaced with values from the GPDK 18 nm process models before using this bound to choose device sizes:

Table 2: Sensitivity of the write bound to the NMOS-to-PMOS strength ratio.

Strength ratio μ_n/μ_p	γ limit from (12)	V_Q at $\gamma = 1$ (V)	V_Q at $\gamma = 0.5$ (V)
2	1.68	0.146	0.067
1.5	1.26	0.211	0.092
1.25	1.05	0.276	0.113
1	0.84	fails	0.146

Reducing γ weakens the pull-up relative to the pass-gate and lowers the written node, but a stronger pass-gate lowers β and loads the bitline, and γ also moves V_M: the final pull-up ratio is not selected by minimizing V_Q alone.

Chosen sizes

Device	**(W/L)**	**Notes**
Pull-down (PD)	$4\times$ fin, $L = 18\,\mathrm{nm}$	$n_{\mathrm{fin}} = 4$
Pull-up (PU)	$2\times$ fin, $L = 18\,\mathrm{nm}$	$n_{\mathrm{fin}} = 2$
Pass-gate (PG)	$2\times$ fin, $L = 18\,\mathrm{nm}$	$n_{\mathrm{fin}} = 2$
β	2	PD/PG = 4/2
γ	1	PU/PG = 2/2

β sweep (schematic, $V_{DD} = 0.8\,\mathrm{V}$)

$\boldsymbol{\beta}$	**$\mathrm{SNM}_{\mathrm{read}}$**	**0-node disturb**	**Source**
$1\times$ (PD 2 fin)	126.6 mV	134.3 mV	`results/part1/read-snm/beta1/snm.json`
$1.5\times$ (PD 3 fin)	145.8 mV	96.6 mV	`results/part1/read-snm/beta1p5/snm.json`
$2\times$ (PD 4 fin)	156.3 mV	77.1 mV	`results/part1/read-snm/beta2/snm.json`

$\mathrm{SNM}_{\mathrm{read}} = \min(\mathrm{SNM}_L, \mathrm{SNM}_R)$ by maximum-square fit on the read butterfly (WL= V_{DD}, BL=BLB= V_{DD}). Method: `scripts/check_snm_sweep.py` splits the 4-column Cadence export to `q.csv`/`qb.csv`, mirrors one VTC, finds three intersections on a 16001-point grid, fits the largest square in each lobe. Convergence checked 4001 vs 16001 ($< 0.05\,\mathrm{mV}$ for $\beta = 1.5, 2$; $0.04\,\mathrm{mV}$ for $\beta = 1$). Cell kept symmetric.

Design point selected: $\beta = 2$, $\gamma = 1$. Cell kept symmetric.

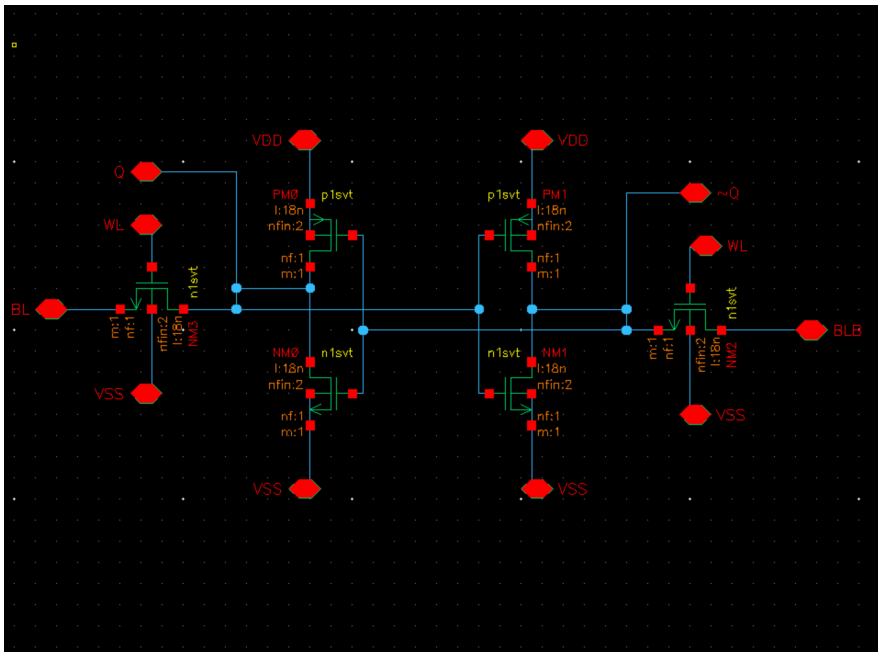


Figure 4: 6T bitcell schematic with transistor sizes (`proj1_lib` 6t-sram, Virtuoso capture).

2. Layout

DRC- and LVS-clean layout: wordline horizontal, bitline pair vertical, tileable for Part 2. Report area in μm^2 and F^2 ($F = 10\,$nm).

Metric	Value
Cell area	*TBD* μm^2
Cell area in F^2	*TBD* F^2
DRC	*clean / summary*
LVS	*clean / summary*

3. Extraction

Post-layout parasitic extraction. Report per-cell bitline-node and wordline-node capacitances (loads for Part 2).

Capacitance	Value
C_{BL} (per cell)	*TBD*
C_{WL} (per cell)	*TBD*

4. Stability characterization (post-layout, $V_{DD} = 0.8\,$V)

4a. Hold SNM

Butterfly plot with $WL = 0$; report $SNM_{\mathrm{hold}} = \min(\mathrm{SNM}_L, \mathrm{SNM}_R)$.

4b. Read SNM

Read butterfly with WL= V_{DD}, BL=BLB= V_{DD} at $V_{DD} = 0.8\,\text{V}$ (schematic). Each β uses two DC sweeps: sweep Q, record $\overline{Q}$; sweep $\overline{Q}$, record Q (testbenches `media/part1/read_snm_q_tb_sch.png`, `media/part1/read_snm_qb_tb_sch.png`; per-β cell schematics `read_snm_q_sch_*.png`, `read_snm_qb_sch_*.png`. The 4-column ViVA export (`results/part1/read-snm/beta*/raw.csv`) is split to `q.csv`/`qb.csv` and evaluated by `scripts/check_snm_sweep.py` (wrapper around `scripts/butterfly_snm.py`): mirror one VTC across $y = x$, interpolate on a uniform 16001-point grid, locate three intersections, fit the largest axis-aligned square in each lobe. Reported $\text{SNM}_{\text{read}} = \min(\text{SNM}_L, \text{SNM}_R)$.

β	SNM_L	SNM_R	SNM_{read}	Intersections (mV)
1	126.6 mV	126.6 mV	126.6 mV	134.3, 410.0, 797.9
1.5	145.8 mV	145.8 mV	145.8 mV	96.6, 388.0, 798.6
2	156.3 mV	156.3 mV	156.3 mV	77.1, 372.5, 798.8

Lobes are symmetric to $< 0.1\,\text{mV}$ (symmetric cell). The low-side intersection is the DC voltage of the "0" node: 134.3 mV ($\beta = 1$), 96.6 mV ($\beta = 1.5$), 77.1 mV ($\beta = 2$), decreasing with β as the Section 1 divider predicts ($V_Q \approx V_{DD}/(1+\beta)$ gives 400/320/267 mV resistive; square-law $(V_{DD} - V_{Tn})(1 - \sqrt{\beta/(1+\beta)})$ is closer but still long-channel only). Full machine-readable results: `results/part1/read-snm/beta*/snm.json` with squares, grid step, and 4k/16k convergence.

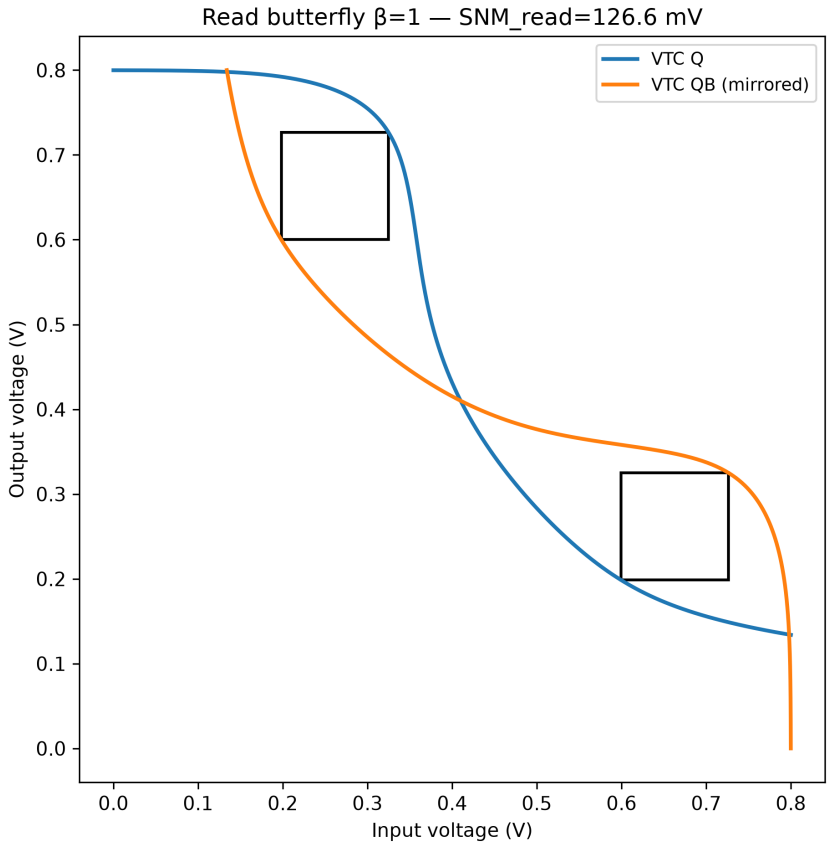


Figure 5: Read butterfly $\beta = 1$: two VTCs + largest squares. $\text{SNM}_{\text{read}} = 126.6\,\text{mV}$. Clean plot from `check_snm_sweep.py`; Cadence screenshot `media/part1/read_snm_beta1.png` in repo.

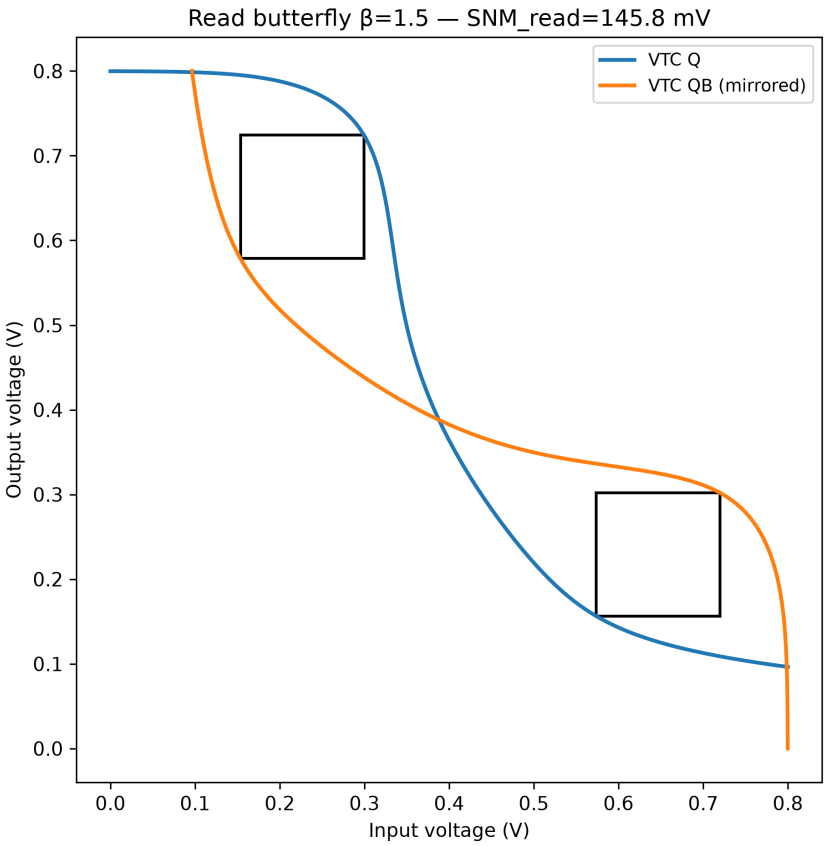


Figure 6: Read butterfly $\beta = 1.5$: $\mathrm{SNM}_{\mathrm{read}} = 145.8\,\mathrm{mV}$.

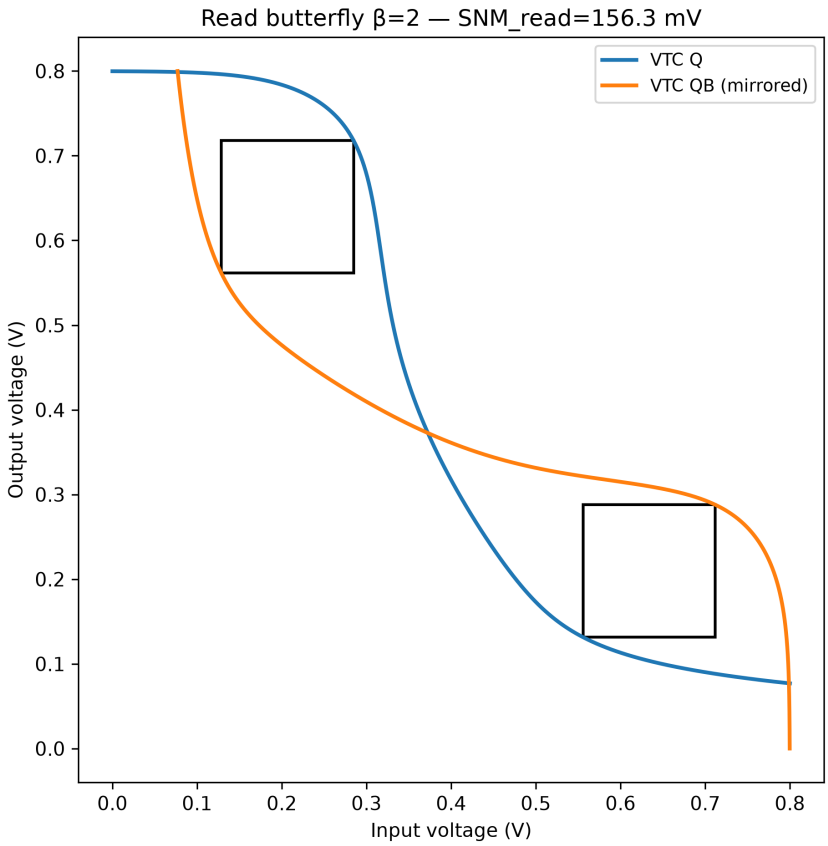


Figure 7: Read butterfly $\beta = 2$ (selected): $\mathrm{SNM}_{\mathrm{read}} = 156.3\,\mathrm{mV}$.

4c. Write margin

Store $Q = 1$, $WL = V_{DD}$, $BL = V_{DD}$; sweep BLB; report $\mathrm{WM}_{\mathrm{BL}} = V_{\mathrm{trip}}$.

4d. Cell current

Report I_{cell} from the "0"-side bitline ($WL = V_{DD}$, bitlines at V_{DD}).

4e. Supply dependence

V_{DD}	$\mathrm{SNM}_{\mathrm{read}}$	$\mathrm{WM}_{\mathrm{BL}}$
0.8 V	156.3 mV ($\beta = 2$ schematic)	*TBD*
0.7 V	*TBD*	*TBD*
0.5 V	*TBD*	*TBD*

Still reads and writes at 0.5 V? *yes/no + notes*.

5. Functional check

Transient test with ideal WL and precharge pulses; bitline pair loaded by 15 additional per-cell bitline loads. Write 0, read; write 1, read. Show WL, BL, BLB, Q, $\overline{Q}$.

6. Discussion

Short paragraph: what limits the cell, and what you would change for Part 2.

Technology note

The assignment specifies the Cadence GPDK 18 nm technology and nominal $V_{DD} = 0.8\,\mathrm{V}$ (Part 1 characterized at 0.8 V with a 0.8/0.7/0.5 V supply sweep; $F = 10\,\mathrm{nm}$ for area in F^2). The Cadence schematic inspected on Betty on October 10, 2026 uses devices labeled with $L = 18\,\mathrm{nm}$, consistent with that GPDK 18 nm flow. All layout, extraction, and post-layout measurements must use that flow. Schematic read-SNM values in §1 and §4b are measured from Spectre DC sweeps (`results/part1/read-snm/beta*/raw.csv` → `snm.json`); all other tables remain TBD until post-layout benches run.